

**0DTE nearest-OTM package: per-rule return summary, entry 12:30 ET, one-bar hold, models compared on the same 866 days**

|                                                      | $n$ | mean   | std   | min   | max  | skew  | ex. kurt | $t$   | Sharpe <sub>ann</sub> | Sharpe <sub>ann</sub> <sup>x</sup> |
|------------------------------------------------------|-----|--------|-------|-------|------|-------|----------|-------|-----------------------|------------------------------------|
| <i>Short volatility (always short)</i>               |     |        |       |       |      |       |          |       |                       |                                    |
| all models (no forecast)                             | 866 | 0.013  | 0.115 | -1.23 | 0.23 | -3.74 | 25.06    | 3.29  | 1.78                  | -2.88                              |
| <i>sign(s) (<math>\pm 1</math> on the sign of s)</i> |     |        |       |       |      |       |          |       |                       |                                    |
| baseline (HAR + calendar OLS)                        | 866 | -0.002 | 0.116 | -1.23 | 0.78 | -0.31 | 22.80    | -0.41 | -0.22                 | -4.86                              |
| block-diagonal ridge                                 | 866 | -0.000 | 0.116 | -1.23 | 0.78 | -0.88 | 22.79    | -0.11 | -0.06                 | -4.70                              |
| block-diagonal ridge, without the FOMC columns       | 866 | 0.001  | 0.116 | -1.23 | 0.78 | -0.82 | 22.82    | 0.15  | 0.08                  | -4.56                              |
| LightGBM <sup>†</sup>                                | 866 | 0.001  | 0.116 | -1.23 | 0.78 | -0.22 | 22.82    | 0.25  | 0.13                  | -4.51                              |
| XGBoost <sup>†</sup>                                 | 866 | -0.000 | 0.116 | -1.23 | 0.78 | -0.26 | 22.80    | -0.07 | -0.04                 | -4.68                              |
| lasso (causally tuned) <sup>†</sup>                  | 866 | -0.003 | 0.116 | -1.23 | 0.78 | -0.78 | 22.76    | -0.65 | -0.35                 | -4.99                              |
| lasso ( $\alpha = 10^{-4}$ )                         | 866 | -0.001 | 0.116 | -1.23 | 0.78 | -0.87 | 22.78    | -0.27 | -0.14                 | -4.78                              |
| elastic net (causally tuned) <sup>†</sup>            | 866 | -0.001 | 0.116 | -1.23 | 0.78 | -0.75 | 22.78    | -0.30 | -0.16                 | -4.79                              |

One trade a day: enter the nearest-OTM package at 12:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. One expiration day = one return; mid fill. Every  $t$  here is the plain  $t = \sqrt{n} \cdot \text{mean}/\text{std}$ , with no autocorrelation correction.

“ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. Sharpe<sub>ann</sub> = (mean/std)  $\times \sqrt{252}$ ; every other column is daily. The one column this table adds to the deck's is Sharpe<sub>ann</sub><sup>x</sup>, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed  $\alpha = 10^{-4}$ ; elastic net (causally tuned). <sup>†</sup> marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is  $s_t = \widehat{RV}_t - IV_{\text{hr}}^2 h_t w_t$ : the forecast against the implied variance of the same window. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining  $h_t$  hours — so every bar carries a signal and no bar sits flat.